Virtual University of Pakistan

BT503

BT101 MIDTERM & FINALTERM MCQS & QUIZ SOLVED

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BT503 MidTerm & FinalTerm MCQs Solved

MIDTERM MCQS

1. **Natural estrogens deactivated by…… Glucuronidation**
2. **Insect virus that causes over expression of genes in molecular biology… baculovirus**
3. **. Acute toxins…… All**
4. **By using recombinant technologies scientists are now producing…… Insulin**
5. **……are combustible materials, containing within themselves all oxygen needful for their combustion, which burn but do not explode, and function by producing gas which produces an explosion…… Propellants or low explosives**
6. **…… are types of waste stabilization pond used for biological treatment of industrial and domestic wastewater…… Facultative lagoon**
7. **PAH-containing fractions, by animal test systems ……evidence have been obtained that PAH containing…….. (skin painting, lung implantation)**
8. **Pentane is utilized as a ……in the preparation of phenol…… foaming**

9. Polystyrene, as a propellant for aerosol sprays, as a filling of low-temperature---------- and as reference compound in gas chromatography. thermometers
10. Herbicide resistant plants ……. pests
11. Average temperature of ocean……. 1–3°C
12. Which of ....does not use to make vessel of biofilters---------may be formic sheet
13. Which of the following is not fate of water ----------may be generation of aerosol
14. Which of the following does not classify as a substitution material.-------- (other than those options) materials is a major potential avenue for the environmental biotechnology
15. Which of the following statement about desulphurization of oil and coal is false--------
16. Metals are released into the environment by all of the following except---------- Earthquake
17. When facultative pond receives raw sewage they are called .....ponds. primary pond
18. Which of the following is not linked to biomagnifications
19. Why frogs are helpful as bioindicators..... Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats
20. For land base application soil type best suited to biotechnological intervention are..... Sand and gravel
21. Attack by insect not only cause damage to the plant but also provide route..... for bacterial or fungal infection
22. Insertion of foreign DNA in the vector resulting in the loss of the future. marker gene
23. Methogenic contortion of over 100 bacterial clone with the capability to digest ... terephthalate
24. Preliminary treatment involves the removal of coarse…… only solids
25. Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. If an indefinite population growth
26. Septic tanks are also a form of,,,,,,,,,, treatment ,primary
27. Mere a gene is one of ……..of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form cluster
28. Facultative ponds which receive raw sewage is also known as ……..primary ponds
29. Facultative ponds which receive raw sewage is also known as primary ponds…..Retention by the plants
30. In the Inhofe tank, settling occurs in the…… upper compartment
31. Second strategy is usually ……..plasmid
32. Fundamental requirements are ……..all of given above

33. Attack by insects is cause……micrococcus lentos
34. Na+/H+ antiport, …. …in opposite directions across a membrane the transport of sodium and hydrogen ions
35. Frog are bio indicator by……. Change in fresh and terrestrial
36. Wastewater is applied to the soil, supplying water and nutrients necessary for-----plant growth.
37. Bio filter consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic
38. The metals released in air by the processes of weathering, erosion and volcanic activity
39. The nature of the soil or water which harbors the pollution can have a major effect on the actual expressed end-result.
40. ----------are microscopic animals living near to the surface of the water body. They are poor swimmers, instead relying on tides and currents as a transport mechanism. They feed upon phytoplanktons, bacterioplanktons, or detritus (i.e. marine snow). Zooplanktons
41. ---------- also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop. Phytoplanktons,
42. Mere a gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reeducates, which converts mercury from an ionic to a volatile form.
43. The insect virus ----------has been shown to be the method of choice for the over expression of genes in many applications of molecular biology. Baculovirus,
44. A strain of ----------putrid known to degrade PCBs Pseudomonas

45. ----------which receive raw sewage is also known as primary ponds. Facultative ponds

46. Bio substitution are: Biofuels, Polymers, Biodegradable

47. septic tank can be a single-chamber tank or a two-compartment tank called an Inhofe tanks
48. In the Imhoff tank, settling occurs in the upper compartment settling compartment

49. In the soil, a pollutant has basically ----------possible destinations: retention in the soil matrix, retention by the plants, appearance in the underground water and collection by underdrains four

50. Facultative ponds which receive raw sewage is also known as primary ponds

51. A strain of Pseudomonas putida known to ----------degrade PCBs

52. Which of does not use to make vessel of biofilters ---------- may be formic sheet
53. Which of the following is not fate of water ……may be generation of aerosol
54. Which of the following does not classify as a substitution material …….....other than those options) materials is a major potential avenue for the environmental biotechnology
55. Which of the following statement about desulphurization of oil and coal is false ……the (Sate
56. Metals are released into the environment by all of the following except …….(Earthquake)
57. When facultative pond receive raw sewage they are called ponds ……. (Primary pond)
58. Which of the following is not linked to biomagnifications …(other than those) • long-lived, • mobile, • soluble in fats, • biologically active.
59. Why frogs are helpful as bioindicator--------- Frogs are also Bioindicators; influenced by changes that take place in their freshwater and terrestrial habitats
60. For land base application soil type best suited to biotechnological intervention are Sand and gravel

61. The septic tank can be a single-chamber tank or a two-compartment tank called an Inhofe tank
62. In the Inhofe tank, settling occurs in the upper compartment settling compartment
63. Attack by insect not only cause damage to the plant but also provide route … for bacterial or fungal infection
64. Insertion of foreign DNA in the vector resulting in the loss of the future ------ --marker gene
65. Methogenic contortion of over 100 bacterial clone with the capability to digest ------terephthalate
66. Preliminary treatment involves the removal of coarse ……solids only
67. Which of does not use to make vessel of biofilters ……may be formic sheet
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69. Which of the following does not classify as a substitution material----------
70. Which of the following statement about desulphirization of oil and coal is false ---------
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72. When facultative pond receive raw sewage they are called ponds ……primary pond
73. Which of the following is not linked to biomagnification ----------

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82. The insect virus, Baculovirus, has been shown to be the method of choice for the over , expression of genes in many applications of ----------molecular bioliogy
83. A strain of Pseudomonas putrid known to---------- degrade PCBs
84. Facultative ponds which receive raw sewage is also known as--------primary ponds
85. Bio substitution are: ……1-Biofuels, 2-Polymers 3-Biodegradable
86. In the soil, a ----------has basically four possible destinations: pollutant
87. -------- is the application of biotechnology for solving environmental problems.
Environmental biotechnology
88. -------- is the degree to which a chemical substance or a particular mixture of substance can damage an organism. Toxicity
89. The presence of contaminents in the environment which produce the serious changes in the envirinment is called ----------. Environmental pollution
90. Herbicide resistant plants ----------
91. Natural oestrogens are deactivated in human by ----------. Glucuronidation
92. Making the compound more polar and easily cleared from the blood by the ----------. Kidney

93. Chlorinated polycyclic aromatic hydrocarbons (CL-PAHs) are group of compound with two or more aromatic rings and one or more --------- atoms are attatched to the ring system. Chlorine
94. ---------- is the presence of contamination in the environment which harm the environment. Pollution
95. Acute toxins ---------- All

BT503 QUIZ-1 SOLVED

1. Biological system may be used to clean up pollutants which of the following attributes must be present in the pollutants. It should be highly toxic
2. Halophiles are extremely important in environment biotechnology these are microorganisms that survive in highly saline environment . they do so by ensuring that-----. Their cytoplasm contains a higher concentration of solutes then their surroundings
3. Glyphosate (herbicide) is analyze of phosphenol pyruvate and inhibits the enzyme, the plants that gave the best result were --------. Petunias
4. Lichens are used as bioindicators to monitor -------- quality. Air
5. Microorganisms are now being used for the removal of pollutants from environment . all of the following statementsin this regards are true except-------. The tolence degradation pathway and p-cumate catabolic pathway are coded by genes present on the same operon.
6. In bacteria genetic transfer of materials occurs through --------. Plasmids
7. Which statement about endocrine disrupters is true------. Feminization of male fish has occurred due to increased levels of oestrogen in water ways,

8. Plants have been produced using the Ti plasmid transfer system of the bacterium. Agrobacterium tumefaciences
9. The plant pathogen that is used to increase the resistance of plant to disease is ----------. Erwinia Carotovora
10. Living organisms may be used to define the characteristics of biosphere , those that are used to get quantitative information on the quality of the environment are known as ---------. Biomonitors
11. The removal of pollutants by plant that breaking down into compounds that can be vaporized is known as -------. Degradation, phytovolatislation, phytosorption
12. Planktons are often used as indicator of water pollution, many planktons reproduce at increased rate in lakes when there are high centralization of ------. Phosphorus and nitrogen

BT503 QUIZ-2 SOLVED

1. Which of the following method for removal of pathogenic organisms do not need chemical products of energy , but require large areas. Maturation ponds
2. Free water in the sludge can be removed by --------. Gravity
3. Which zone is essential in the treatment line for biological removal of phosphorus, aerobic or anerobic. Both
4. A process In sludge management for preparation for dewatering is known as ---------. Dewatering
5. Vitrification uses --------- temperature for fuse contaminated material. High
6. The sludge production is a function of the waste water treatment system used for the ---------. Liquid phase

7. --------- is very effective agent for the removal of pathogens. Ozone
8. Lime stabilization is used in treat ---------- sludge. All
9. Thermal decomposition process by oxidation is known as ------. Incineration
10. For which of the following waste water treatment system the equipement depends on the post treatment process used. Rapid infiltration
11. Which of the following process does not introduce chemical products into the liquid. Membranes
12. A sludge with a level of dry solids of 2% has water content of ------. 98%
13. Which of the following method for pathogen removal affects the reproduction of pathogenic agents. UV Radiations
14. Gamma radiations are used for ----------. Conditioning
15. Thickening in sludge management is removal of -------. Water
16. Most significant temperature for bio-remediations is --------- 20-30℃

BT503 FINAL TERM MCQS

1- Herbicide resistant plants ----------
2- Natural oestrogen deactivated by ---------- glucuronidation
3- Average temperature of ocean ----------
4- Insect virus that causes over expression of genes in molecular biology bacilo virus
5- Acute toxins…… all

6. By using recombinant technologies scientists are now producing……insulin

7.……are combustible materials, containing within themselves all oxygen needful for their combustion, which burn but do not explode, and function by producing gas which produces an explosion…… Propellants or low explosives

8- …… are a type of waste stabilization pond used for biological treatment of industrial and domestic wastewater…… Facultative lagoon

9- PAH-containing fractions, by animal test systems ……evidence has beenobtained that PAH containing…….. (skin painting, lung implantation)

10-Pentane is utilized as a ……in the preparation of phenolic…… foaming

11-Polystyrene, as a propellant for aerosol sprays, as a filling of low-temperature----------, and as reference compound in gas chromatography. thermometers

12-Reintroduction of these ----------bacteria to the polluted sites should give them an advantage over the indigenous bacteria as they would be better suited to survive and remediate the contamination. trained

13-The DNA construct contains ----------which are complementary to the plant DNA to enable the inserted piece to recombine into the plant genome. regions of DNA

14-Attack by insects not only causes damage to the plant but also provides a route for bacterial or fungal infection.

15-Bacteria which Thrive at temperatures above 45°C is called as ------Thermophiles

16-Bacteria which Thrive at temperatures above ~85°C. is called as Hyper thermophiles

17.Which attribute of pollutant is not linked to bio magnification long-lived, • mobile, • soluble in fats, • biologically, active other than these option

18-Following substances cannot use to make the vessel of bio filter concrete, metal or durable plastic, (other than these)

19-Following cannot be adopted to increase the efficiency of phosphorus removal by activated sludge method ?????

20-Following is true about imhoff tank used in anaerobic filter system (see in midterm)

21-When facultative ponds receive raw sewage, they are also called primary ponds.

22- In general, enzymatic biosensors are based on ----------enzymes by different classes of compounds. the selective inhibition of specific

23. Which enzyme are used in leather industry Traditionally, pancreatic enzymes were employed

24. Explosives of characterized by presence of _____group??????

25. Which statement about chlorinated hydrocarbons is correct — fixation of carbon by phytoplankton.

26. Main use of DDT other than fighting crop insects effective in controlling the Anopheles mosquito — malaria

24- The term "endophytes" is usually used to denote non-pathogenic microorganisms.

28. The steps of solid waste management are listed below what is correct ----------

29. Statement about phytoremediation is false

30- State about sampling for situ monitoring is true

31. Following property of hydrocarbon does not contribute poor biodegradation

32. Toxicity represents the potential damage to life and can be both short and long term. It is related to Type of the pollutant, • Concentration,• The time of

33- In order for bio magnification to occur, the pollutant must be:• Long-lived,• Mobile,• Soluble in fats,• Biologically active.

34- bio filters can reduce odour release by 95% or more

35- For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: Addition of coagulants

36- In the Inhofe tank, settling occurs in the…… upper compartment

37- In the soil, a pollutant has basically ----------possible destinations: four

38- Facultative ponds which receive raw sewage is also known as ……..primary ponds

sign of bright future

39- The bio scrubber is not itself truly a biological treatment system, but rather a highly efficient method of removing odour components by dissolving them.

40- In the Imhoff tank, settling occurs in the upper compartment settling compartment).

41-. Biodiesel is derived mostly from---- vegetable oil

42- DDT was of great benefit to humans in their fight against--- ---------- malaria

43- Two characteristics of chlorinated pesticides that led to great harm are

Their resistance to biodegradation and• their hydrophobicity

44- A characteristic of explosives is the presence of ---------- groups. Nitro (-NO2)

45-Endophyte is micro pathogen----------

46- up flow anaerobic sludge blanket reactor stand for UASB

47-Water treatment accounts for 25% of the total global environmental market.

48-Basic equipment required in Preliminary treatment involves screen, grit chamber and flow meter

49-UASB stands for ----------up flow anaerobic sludge blanket reactor

50-Important factors to be considered when evaluating and selecting unit operations and processes except: Sterilization of vessels

51- What is false about final disposal of sludge in ocean? It has beneficial uses.

51-Rhizodegradation refers to the biodegradation of contaminants in the soil by edaphic microbes enhanced by the inherent character of the rhizosphere itself.

53-False about On-site disposal? waste volumes are large

54. . What is true about Biodiesel? It is derived mostly from vegetable oils

55. The term "endophytes" is usually used to denote non-pathogenic microorganisms.

56. A tumor-like growth, seen as a crown gall in plants. Agrobacterium tumefaciens

57. Dioxins released as by-products in a number of chemical processes Do not pose threats to environment

58. Explosives use ----------functional group nitro

59. Aerobic Co-metabolism of ----------NADH to provide the 2H CAHs-II requires

60. What is false about poor biodegradation of hydrocarbons? Easily degraded by enzymatic attack

61. Nonspecific enzymes are oxygenases that initiate the oxidation of hydrocarbons such as methane or toluene.

62. Western blotting is used to detect---------- proteins

63. Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. If an indefinite population growth

were allowed…-------following problems arised except option D will be correct option

64. Wastewater is applied to the soil, supplying water and nutrients necessary for plant growth. Which is not true? Generation of aerosols

65-Which statement is not true regarding desulphurization of oil and coal? D

66.The metals release in air by all following process except-.--------- Earthquake

67. mer A gene codes for the enzyme, ---------- mercuric ion reductase

68-What is true regarding zooplankton? Rely on tides and currents as a transport

69-Facultative ponds which receive raw sewage is also known as primary ponds

70- In the soil, a pollutant has all the possible destinations except? Option d will Be correct

71. What is not true about bio scrubbers? Remove toxic metals

72. One such recombinant is a plant where the fatty acid composition in the seed has been modified to produce triacylglycerol's containing elevated levels of trierucinic acid suitable for use in the polymer industry. Arabidopsis thaliana

73. Unlike other marines species, sharks are characteristically untroubled by ----------and studies of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms. bio fouling

74.) Factors related to bio magnifications except ----------source

REGARDS: SOJHLA FATIMA